

## TAMPA INTL, FL, USA

WMO: 722110

Lat: 27.962N

Lon: 82.540W

Elev: 6

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12842

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |             |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|-------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|-----------|--------------|
|               |            |            | 99.6%                         |            |            | 99%         |            |             | 0.4%                  |             | 1%         |             |                       |           |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP          | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>4.3 | (c)<br>6.5 | (d)<br>-4.9                   | (e)<br>2.5 | (f)<br>8.7 | (g)<br>-2.3 | (h)<br>3.1 | (i)<br>10.1 | (j)<br>9.3            | (k)<br>15.4 | (l)<br>8.4 | (m)<br>15.3 | (n)<br>2.9            | (o)<br>10 | (p)<br>0.356 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>7.1             | (c)<br>33.6     | (d)<br>24.9 | (e)<br>33.0 | (f)<br>25.0 | (g)<br>32.4 | (h)<br>25.0 | (i)<br>26.9         | (j)<br>31.0 | (k)<br>26.6 | (l)<br>30.8 | (m)<br>26.2 | (n)<br>30.5 | (o)<br>3.7           | (p)<br>260 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.8                            | 21.1 | 29.4 | 25.3 | 20.4 | 29.0 | 24.9 | 20.0 | 28.8 | 84.5          | 31.1 | 82.5 | 30.8 | 81.0 | 30.4 | 30.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 8.1               | 7.1  | 6.0 | 0.5                        | 35.2 | 2.2                | 0.7 | -1.1                                               | 35.7 | -2.4       | 36.1 | -3.6       | 36.6 | -5.3       | 37.1 |      |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |      |
|                   |      |     | WB                         | -2.1 | 27.9               | 2.1 | 0.8                                                | -3.6 | 28.5       | -4.9 | 29.0       | -6.1 | 29.4       | -7.6 | 30.0 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 23.3   | 16.1  | 17.9  | 20.0  | 23.0  | 26.1  | 27.9  | 28.3  | 28.5  | 27.8  | 24.8  | 20.6  | 18.1 |
|                                                                                  | DBStd       | 5.33   | 4.56  | 4.25  | 3.71  | 2.70  | 1.93  | 1.43  | 1.21  | 1.18  | 1.39  | 2.85  | 3.51  | 4.46 |
|                                                                                  | HDD10.0     | 13     | 8     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3    |
|                                                                                  | HDD18.3     | 267    | 98    | 55    | 28    | 4     | 0     | 0     | 0     | 0     | 0     | 2     | 19    | 61   |
|                                                                                  | CDD10.0     | 4862   | 196   | 223   | 309   | 389   | 499   | 537   | 568   | 574   | 533   | 460   | 319   | 255  |
|                                                                                  | CDD18.3     | 2074   | 28    | 42    | 79    | 143   | 241   | 287   | 310   | 316   | 283   | 204   | 88    | 54   |
|                                                                                  | CDH23.3     | 18951  | 91    | 164   | 404   | 965   | 2202  | 3019  | 3421  | 3498  | 2854  | 1644  | 473   | 216  |
|                                                                                  | CDH26.7     | 6618   | 4     | 16    | 53    | 224   | 782   | 1183  | 1346  | 1398  | 1051  | 476   | 69    | 17   |
| Wind                                                                             | WSAvg       | 3.0    | 3.1   | 3.2   | 3.5   | 3.4   | 3.1   | 2.9   | 2.5   | 2.5   | 2.7   | 2.9   | 3.0   | 2.9  |
| Precipitation                                                                    | PrecAvg     | 1186   | 58    | 73    | 75    | 44    | 72    | 159   | 187   | 209   | 159   | 54    | 38    | 58   |
|                                                                                  | PrecMax     | 1720   | 160   | 275   | 305   | 272   | 448   | 474   | 523   | 419   | 388   | 158   | 155   | 396  |
|                                                                                  | PrecMin     | 758    | 3     | 7     | 0     | 0     | 0     | 37    | 42    | 72    | 20    | 0     | 0     | 2    |
|                                                                                  | PrecStd     | 253    | 39    | 56    | 66    | 42    | 77    | 88    | 83    | 79    | 86    | 38    | 36    | 72   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 27.3  | 28.5  | 29.4  | 31.3  | 34.0  | 34.5  | 34.1  | 34.2  | 33.9  | 32.5  | 29.8  | 28.4 |
|                                                                                  |             | MCWB   | 21.0  | 21.6  | 20.5  | 21.7  | 23.0  | 24.9  | 25.1  | 25.5  | 24.7  | 24.3  | 22.8  | 22.0 |
|                                                                                  | 2%          | DB     | 25.6  | 26.8  | 28.1  | 30.1  | 32.4  | 33.3  | 33.1  | 33.3  | 32.9  | 31.3  | 28.5  | 27.0 |
|                                                                                  |             | MCWB   | 20.0  | 20.7  | 20.6  | 21.6  | 23.0  | 24.8  | 25.4  | 25.6  | 24.6  | 23.9  | 22.1  | 21.4 |
|                                                                                  | 5%          | DB     | 24.2  | 25.4  | 27.0  | 29.1  | 31.4  | 32.4  | 32.4  | 32.6  | 32.1  | 30.3  | 27.3  | 25.8 |
|                                                                                  |             | MCWB   | 19.6  | 20.0  | 20.4  | 21.4  | 22.9  | 24.9  | 25.5  | 25.6  | 24.7  | 23.5  | 21.4  | 20.9 |
|                                                                                  | 10%         | DB     | 22.8  | 24.0  | 25.8  | 28.0  | 30.5  | 31.6  | 31.7  | 31.9  | 31.3  | 29.3  | 26.1  | 24.4 |
|                                                                                  |             | MCWB   | 18.8  | 19.3  | 19.9  | 21.0  | 22.6  | 24.8  | 25.4  | 25.7  | 24.7  | 23.0  | 20.7  | 20.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 22.6  | 22.8  | 24.0  | 24.6  | 25.7  | 27.0  | 27.5  | 27.8  | 27.1  | 26.3  | 24.6  | 23.6 |
|                                                                                  |             | MCDB   | 25.3  | 26.5  | 26.9  | 28.5  | 30.4  | 31.5  | 31.1  | 31.9  | 30.8  | 29.6  | 27.5  | 26.1 |
|                                                                                  | 2%          | WB     | 21.5  | 21.9  | 22.9  | 23.8  | 25.0  | 26.4  | 26.8  | 26.9  | 26.5  | 25.6  | 23.8  | 22.8 |
|                                                                                  |             | MCDB   | 23.9  | 25.0  | 25.7  | 27.6  | 29.6  | 31.0  | 30.9  | 31.2  | 30.5  | 29.0  | 26.6  | 25.1 |
|                                                                                  | 5%          | WB     | 20.5  | 21.2  | 22.1  | 23.0  | 24.5  | 25.9  | 26.4  | 26.6  | 26.0  | 25.0  | 22.9  | 22.0 |
|                                                                                  |             | MCDB   | 22.9  | 24.0  | 24.8  | 26.9  | 28.9  | 30.4  | 30.7  | 30.9  | 29.9  | 28.4  | 25.7  | 24.3 |
|                                                                                  | 10%         | WB     | 19.5  | 20.4  | 21.2  | 22.4  | 24.0  | 25.5  | 25.9  | 26.1  | 25.6  | 24.5  | 22.0  | 21.1 |
|                                                                                  |             | MCDB   | 22.1  | 23.0  | 24.1  | 26.2  | 28.3  | 30.0  | 30.1  | 30.4  | 29.4  | 27.7  | 24.7  | 23.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.6   | 9.4   | 9.4   | 9.1   | 8.9   | 7.5   | 7.0   | 7.1   | 7.3   | 8.3   | 9.3   | 9.0  |
|                                                                                  |             | MCDBR  | 9.3   | 9.3   | 9.6   | 9.6   | 9.6   | 8.4   | 7.7   | 7.9   | 8.2   | 8.2   | 9.0   | 8.4  |
|                                                                                  |             | MCWBR  | 5.0   | 4.7   | 4.3   | 4.0   | 3.5   | 3.2   | 3.1   | 3.0   | 2.8   | 3.1   | 4.2   | 4.2  |
|                                                                                  | 5% WB       | MCDBR  | 7.9   | 8.0   | 7.7   | 8.2   | 8.1   | 7.3   | 7.0   | 7.1   | 7.0   | 6.9   | 7.5   | 7.2  |
|                                                                                  |             | MCWBR  | 5.2   | 4.7   | 4.9   | 4.4   | 3.6   | 3.3   | 3.4   | 3.3   | 3.1   | 3.2   | 4.6   | 4.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.343  | 0.355 | 0.368 | 0.392 | 0.425 | 0.450 | 0.474 | 0.455 | 0.422 | 0.390 | 0.354 | 0.342 |      |
|                                                                                  | taud        | 2.520  | 2.492 | 2.444 | 2.343 | 2.309 | 2.316 | 2.254 | 2.321 | 2.383 | 2.456 | 2.501 | 2.548 |      |
|                                                                                  | Ebn at Noon | 902    | 918   | 923   | 905   | 870   | 843   | 822   | 837   | 858   | 869   | 881   | 887   |      |
|                                                                                  | Edh at Noon | 91     | 101   | 112   | 127   | 132   | 130   | 138   | 129   | 117   | 103   | 92    | 85    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.54   | 4.27  | 5.40  | 6.48  | 6.87  | 6.19  | 5.86  | 5.64  | 5.14  | 4.74  | 3.86  | 3.27  |      |
|                                                                                  | RadStd      | 0.25   | 0.28  | 0.45  | 0.32  | 0.38  | 0.50  | 0.36  | 0.35  | 0.34  | 0.29  | 0.20  | 0.27  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.53 | +1.07   | N/A    | +0.20   | N/A   | -0.25 | -5          | -51     | +185    | +125    |
| (47) Regional (38 neighbors) |  | +0.42 | N/A     | N/A    | N/A     | N/A   | N/A   | -4          | -47     | +164    | +116    |

Nomenclature: See separate page